# AI Image Generator

## 13. Change Log and Roadmap

Version 2.0
22 October 2026

Project owner: Frazile AI
Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Z-Image Turbo, Stable Diffusion XL and its best fine-tunes, FLUX.1, LoRAs and Real-ESRGAN, served locally with Gradio in the Frazile AI studio UI, with per-image GPU telemetry recorded in SQLite and a live Admin panel.

# 13. Change Log and Roadmap

## 13.1 Change log

### Increment 1: command-line generator (21 Oct 2026)

- Created the project folder, virtual environment, and requirements.txt; installed torch 2.14.1+cu126, diffusers 0.41, transformers, accelerate.
- Wrote generate.py with prompt, negative, steps, guidance, size, seed, count and model arguments. First image: 1024x1024 in 9.7 s, peak VRAM 10.5 GB.

### Increment 2: web interface

- Added app.py using Gradio 6.30 with a gallery and an info box; model loaded once at startup and kept resident. Added run-app.bat and README.

### Increment 3: prompt-driven controls

- Removed style, size, steps and guidance controls. Added prompt_parser.py, the live "Will generate" line and a collapsible help table.

### Increment 4: telemetry and history

- Added gpu_info.py and db.py with a 45-column SQLite table; per-image timing and GPU sampling; the History tab.

### Increment 5: real 4K and 8K output

- Added upscaler.py (Real-ESRGAN x4plus, tiled, multi-pass); parser upscale words; eight upscale columns; 4K in 3.4 s, 8K in 8.5 s.

### Documentation 1.0 and repository (22 Oct 2026, 09:17)

- Documentation set published as PDFs under Frazile AI ownership. Project pushed to github.com/Fzhacking/ai-image-gen (private).

### Increment 6: model registry (22 Oct 2026, 10:09)

- Added models.py: SDXL base, Juggernaut XL v9, RealVisXL V5.0, DreamShaper XL v2 Turbo, Playground v2.5, Animagine XL 3.1 and FLUX.1 schnell; local checkpoints; LoRAs; compel weighting; per-model samplers and defaults; ModelManager with verified VRAM release.
- Parser: model and LoRA directives, explicit-flag tracking, apply_defaults(). App: Model dropdown with Auto, model in details and history. CLI: --model, --lora, --list-models.
- Installed peft 0.21, compel 2.5, bitsandbytes 0.50, sentencepiece, protobuf. Downloaded the five fine-tunes (32.5 GB).
- Fixed: compel padding for SDXL dual encoders; comma tidy-up; FLUX fidelity note.
- Measured: every SDXL-family model at 832x1216 in 2.0 to 10.5 s, peak 13.2 GB, swap 26 s.

### Increment 7: Frazile AI branding and Upscale bar (22 Oct 2026, 10:27)

- Dark studio theme with pink-to-yellow accents, applied to both colour modes; branded header; Inter and JetBrains Mono.
- Upscale bar after generation: 2K / 4K / 8K, selected image or whole batch, records updated in place with the new db.update().

### Increment 8: full width, Admin panel, Docker (22 Oct 2026, 10:50)

- Full-width layout, version pills, Clear button, "Loaded now" line.

- Admin tab with nine tiles and six sections; sysinfo.py collector on psutil and nvidia-smi; 5 s auto-refresh; optional reachability check; caching so a refresh costs 0.7 s.
- Dockerfile, docker-compose.yml and .dockerignore; app reads bind address, port and browser flag from the environment.

### Increment 9: Z-Image Turbo as the default model (22 Oct 2026)

- Added the zimage family to models.py with a bf16 loader and CPU offload under 32 GB; Z-Image Turbo (Tongyi-MAI, 6B, 8 steps, Apache 2.0) is the model loaded at start-up.
- The repository ships the transformer in fp32 (23 GB); the first load converts it to bf16 and saves a 12 GB copy under models/cache/ so later loads take about about 130 s. The Qwen3 text encoder is loaded in 8-bit; everything stays on the GPU (15.8 GB resident, 18.1 GB peak).
- Measured: 1344x768 in 9.8 s and 832x1216 in 8.6 s at 8 steps; the FRAZILE AI neon-sign test rendered the text correctly.
- Auto no longer switches away from Z-Image or FLUX; style-based specialist selection applies only while an SDXL-family model is loaded.
- Fixed: CPU offload of Z-Image crashed the process on the 16 GB machine (fp32 shards plus offload exceeded system RAM); replaced by shard-by-shard loading straight onto the GPU.

### Documentation 2.0 (22 Oct 2026)

- This document set, rebuilt from docs-src/build_pdf.py with three new sections (Models, Admin panel, Deployment) and a model comparison figure.

## 13.2 Roadmap

| Priority | Item | Notes |
|---|---|---|
| High | Image-to-image and inpainting | Same pipeline family; lets a chosen image be edited rather than regenerated; the ModelManager already shares components |
| High | Character consistency with IP-Adapter or reference images | Keeps a subject's identity across prompts; the usual next step after LoRAs |
| High | Test the Docker image on a host with WSL 2 and Docker Desktop | Written and reviewed but unbuilt on the development machine |
| Medium | FLUX.1 dev and ControlNet | dev is gated and non-commercial; ControlNet gives pose and depth control |
| Medium | SDXL refiner or hires-fix pass | Adds generated detail at 2x before Real-ESRGAN |
| Medium | Export history to CSV and chart timings per model | pandas is already present |
| Medium | Thumbnail cache for the History grid | Avoids loading full PNGs when browsing |
| Low | Authentication | Gradio supports basic auth; needed before exposing beyond localhost or publishing the container port |
| Low | Cloud deployment profile | The Docker image is the starting point for RunPod or a Hugging Face Space |